IIA WEB DESIGN • UNIT 12

Website Hosting and Publishing

Domains • DNS • hosting choices • GitHub Pages • safe publication

Bachelor of Library and Information Studies | Bachelor of Records Management Systems

Learning outcomes

Understand publishing

Explain hosting, servers, domain names, DNS, HTTPS and the basic steps used to publish a static webpage.

Choose and protect

Compare common hosting models and use simple checks for usability, accurate information, privacy and ongoing maintenance.

KEY IDEA By the end, you should know what it takes to make a useful information-service website available to visitors.

Why publishing matters

Before publishing

An HTML file saved on one computer may display correctly in its browser but cannot automatically be opened by someone elsewhere.

After publishing

A hosting service makes the website available through an Internet address, where visitors can reach its public pages.

KEY IDEA Publishing turns a collection of local files into a reachable web service.

Local webpage versus published website

Local webpage

The address might start with file:// and refer to a document stored on one computer. Such a path is not a public website address.

Published webpage

A visitor opens an https:// address; the browser requests content from a public hosting service and displays the response.

KEY IDEA A webpage can work locally even when links or resources will fail after deployment.

What does web hosting mean?

Hosting service

Stores website files or runs applications connected to the Internet, so people can visit them using a browser.

Hosting responsibilities

May include file delivery, configuration, storage and other capabilities, depending on the provider and plan.

KEY IDEA Hosting is the service; a domain name is an address used to reach it.

What is a web server?

A request arrives

When a person opens a web address, the browser sends a request to the appropriate service.

A response is returned

A web server can deliver HTML, CSS, JavaScript, images or results produced by an application.

KEY IDEA Web servers help deliver webpages; the browser renders the content for the visitor.

Deployment: moving from design to delivery

Preparation

Check index.html, supporting files, links and content before making the site available to visitors.

Deployment

Place the files or application on a suitable hosting platform and test them using the published address.

KEY IDEA Publishing should include verification, not merely copying files.

Understand a complete URL

Example

https://example.org/services.html identifies a resource using the HTTPS scheme, host and page path.

Why it matters

Visitors need correct URLs and clear navigation to find public services, catalogues or request guidance.

KEY IDEA A URL identifies a resource; the domain is only one part of its address.

Domain names: readable online names

What a domain does

A name such as example.org is easier to remember and share than a numerical network address.

What a domain does not do

Registering a domain does not automatically create a website or provide a database and hosting.

KEY IDEA Domain registration and hosting are related but distinct services.

DNS: how a name finds a service

Domain Name System

DNS helps the browser identify where to connect when a user enters a domain name.

Typical records

An A record provides an IPv4 address; a CNAME record associates one hostname with another hostname.

KEY IDEA DNS records must match the provider’s instructions for the site to resolve correctly.

Provider address and custom domain

Provider-supplied address

A hosted site may receive a URL such as username.github.io/project/ without purchasing a separate domain.

Custom domain

An institution may use its own registered name, with DNS configured to direct visitors to its hosting service.

KEY IDEA A beginner can publish an example site without buying a custom domain.

HTTP and HTTPS

HTTP

A protocol for requesting and transferring web resources between a browser and a server.

HTTPS

Uses TLS to encrypt information during transport and authenticate the connection to the website endpoint.

KEY IDEA Use HTTPS for public sites, especially where visitors need a trusted connection.

What HTTPS does not guarantee

Connection protection

HTTPS helps keep data from being read or changed in transit by an ordinary network observer.

Not full system security

It does not by itself verify an organisation’s honesty, protect a database or enforce staff access permissions.

KEY IDEA Security also needs correct authentication, authorisation, safe design and operational safeguards.

Three common hosting approaches

Static hosting

Serves HTML, CSS, JavaScript and images. Suitable for public guides, simple service pages and portfolios.

Managed content or application hosting

CMS and application hosting can support content editing or complex back-end processes with appropriate administration.

KEY IDEA Hosting choice follows the website’s required functions, users and data.

What can a static site do?

Typical features

Show opening hours, services, directions, contact details and links to separate catalogue or records systems.

Limitations

Static files alone do not manage secure user accounts, confidential record workflows or server-side databases.

KEY IDEA A static page can explain a service without being the protected system that operates it.

When a content management system helps

Content editing

A CMS lets authorised users create, revise and publish webpages using an administration interface.

Example

Library staff update service notices and announcements without opening and editing individual HTML files.

KEY IDEA A CMS assists content maintenance; it is not automatically a library catalogue or records system.

When specialist application hosting is needed

Institutional functions

Live catalogue availability, restricted records access, audit trails and retention workflows need appropriate application services.

Operational requirements

Reliable hosting, access control, backups, patching and staff support are essential to the service design.

KEY IDEA An information-service application is more than an attractive HTML form.

Compare free and paid hosting

Free options

Useful for coursework and uncomplicated public static sites, subject to provider restrictions and usage limits.

Paid options

May offer more storage, support or features, but ongoing fees and security responsibilities need checking.

KEY IDEA Avoid judging by price alone; compare purpose, limits, ownership, reliability and support.

Questions to ask a hosting provider

Function and limits

Which site features work? Are bandwidth, file size, traffic and custom domains restricted?

Support and continuity

How are backups, domain renewal, account access, availability and recovery handled?

KEY IDEA A provider’s current published terms are more dependable than an old tutorial.

Meet GitHub and GitHub Pages

GitHub repository

A place to store and track website source files. Collaborators can review changes according to permissions.

GitHub Pages

Publishes static website content from a configured repository source or workflow.

KEY IDEA GitHub Pages is one publishing option—not a secure records-management backend.

Prepare a GitHub Pages website

Entry file

A simple website should include index.html at the selected publishing root.

Supporting folders

Keep CSS, images and other pages in sensible directories, with paths that match the files exactly.

KEY IDEA Organised files make navigation and publishing problems easier to detect.

Example website folder structure

Homepage and content

index.html can link to services.html and other HTML pages in the same directory.

Style and media

A folder such as css/ may contain style.css; images/ can hold photographs with appropriate filenames and alt text.

KEY IDEA File paths should stay relative to the correct webpage location.

Publish from a repository branch

Step 1

Place the website files in a GitHub repository that meets your account and Pages publishing requirements.

Step 2

In Settings → Pages, choose Deploy from a branch and select the branch and root or docs folder containing the site.

KEY IDEA After deployment, open the real published URL and test the result.

Understand GitHub Pages site URLs

Typical pattern

A repository site often uses https://username.github.io/repository-name/.

Practical consequence

A starting slash in a link can point to the account domain root rather than your project directory.

KEY IDEA Use carefully constructed relative paths and always check links in the published site.

A private repository is not a private site

Important distinction

Repository permissions control source access, but GitHub Pages sites are generally public even when published from an eligible private repository.

Good practice

Do not include student passwords, borrower details, private records or other restricted information in public website files.

KEY IDEA A concealed URL is not access control; protect restricted services using approved authentication and authorisation.

Common publishing mistake: broken paths

Incorrect reference

An image may load locally from a computer-specific C: drive path but fail for visitors.

Correct approach

Include the file in the website folders and link to it with a relative path such as images/library.jpg.

KEY IDEA A working local preview is not enough: test the online file paths.

Other platforms for static websites

Cloudflare Pages

Supports publishing static web projects through its documented workflow and linked repository options.

Netlify

Supports deployment of basic static webpages through supported site-deployment methods.

KEY IDEA Platform capabilities and free-plan terms can change; compare official documentation.

Institutional example: library website

Public content

Hours, service policies, events and contact details can be served through a static or CMS-supported website.

Specialist services

A separate ILS or repository provides catalogue searches, current item availability or approved digital resources.

KEY IDEA Use links and consistent language to connect public pages to appropriate managed systems.

Institutional example: records centre

Public guidance

A simple webpage may explain enquiry procedures, office location and types of publicly available records.

Restricted processes

Personnel records, authorisations and audit history belong in secure managed records systems.

KEY IDEA Never publish sensitive registry files simply because a public webpage is convenient.

Prepare for visitors using a phone

Responsive layout

Text and navigation should remain readable and functional without horizontal scrolling on narrow screens.

Testing

Check headings, tables, images, forms, tap targets and navigation on both smaller and larger views.

KEY IDEA A webpage is ready for readers only when its key tasks work across common devices.

Accessibility checks before publishing

Readable and operable

Use semantic headings, descriptive links, visible focus, usable keyboard controls and sufficient contrast.

Non-text information

Give images appropriate alternative text, except where a purely decorative image should be ignored.

KEY IDEA Accessibility is part of good service design, not an optional finishing touch.

Copyright and ethical publishing

Permission matters

Use your own photos, properly licensed images or resources you have permission to publish.

Information responsibility

Confirm that descriptions, procedures and records are suitable for public access and do not disclose private data.

KEY IDEA Public availability does not mean every piece of information may lawfully be republished.

Maintain accurate public information

Content reviews

Update library opening hours, contact details and records request guidance when services change.

Technical checks

Repair broken links, review account permissions and ensure site software remains supported where applicable.

KEY IDEA A website that is online but inaccurate can mislead users.

Backup, recovery and account ownership

Continuity

Keep a recoverable copy of source content and know who can restore the site when errors occur.

Governance

Agree who controls hosting credentials, the domain registration and the authority to approve published changes.

KEY IDEA Reliable information services depend on ownership, maintenance and a recovery plan.

Practice: choose a hosting model

Public guide or student project

For a small HTML/CSS website, static hosting is often a sensible starting point.

Controlled internal workflows

An ERMS or catalogue with live transactions needs specialist software and secure application hosting.

KEY IDEA Match the hosting approach to the task and the sensitivity of information.

Practice: publish-readiness checklist

Content and paths

Check the homepage, file references, mobile readability, clear navigation and current contact details.

Safety and rights

Verify public permission for materials and exclude passwords and confidential records.

KEY IDEA A classroom checklist helps with review but is not a substitute for institutional approval.

Assignment 12: a short publishing plan

Choose one scenario

A fictional library public-information site or records-centre public-guidance site.

Explain in 100–150 words

Recommend a suitable hosting approach; state one benefit, one limitation and three checks before publication.

KEY IDEA No account registration, paid hosting or live deployment is required for the assignment.

Review: connect the key terms

Name the essentials

Hosting, server, domain name, DNS, URL, HTTPS, static website and deployment are related but different.

Connect to practice

Choose safe publishing for public information and controlled software for confidential or transactional institutional records.

KEY IDEA Web design knowledge includes understanding where a site lives and who can safely access it.

Unit 12: takeaways and next steps

The full journey

Plan a website → organise HTML and CSS → test the pages → publish safely → check and maintain the live service.

Professional responsibility

Future information professionals help ensure that published content is accurate, accessible, appropriately restricted and maintained.

KEY IDEA You can now bring together design, HTML, CSS, interaction, usability and publishing decisions.